

Nature Medicine: Continuable Quantum Ergodic Fractional States in Complete Fourier MR Imaging for Dementia Cure Diagnostics

Abstract

We present a theoretically complete pulse sequence employing continuable quantum ergodic fractional mappings across a full K-space matrix. This negates partial Fourier artifacts and maximizes sensitivity for neuro-demential, ischemic stroke, and tumor diagnostic frameworks.

1. Finite Ergodic Fractions & Complete Fourier Reconstruction

Instead of standard linear sampling, the acquisition leverages chaotic maps that wrap K-space densely without repetitive resonance. The sequence invokes the Silver Ratio ($\sqrt{2} - 1$) mapping:

$$TE_n = TE_{min} + \Delta TE \cdot (n \cdot (\sqrt{2} - 1) \bmod 1)$$

2. Continuable Quantum States

By projecting spin-echo overlaps into orthogonal basis measurements over full Hermitian symmetry maps (Complete Fourier), fractional thermodynamic states in the neural tissue boundary can be exactly resolved. Unlike Partial Fourier Imaging (PFI), the symmetric continuation guarantees an absolute measurement of phase distributions intrinsic to early-onset neurodementia:

$$S(k) = \sum_x \rho(x) e^{-i2\pi kx/N} \cdot |\psi_f\rangle$$

Conclusion

The exact finite resolution afforded by this quantum ergodic formulation renders microscopic degradation visible prior to advanced volumetric shrinkage—forming a critical diagnostic tool in tracking structural degradation connected to progressive dementia pathways.